



Cryptography

Project 1

Secure Wireless Messaging for IoT Devices

Student(s) _____

Date _____

1. Project Description

The objective of this project is to design and implement a wireless communication system that allows at least two IoT devices to exchange short text messages securely. Each team may select the communication technology, message format, and cryptographic mechanisms according to the capabilities of the selected hardware. The wireless channel must be considered untrusted, and the security of the system must therefore rely on the cryptographic protocol rather than on the communication medium.

The final system must provide confidentiality, integrity, device authentication, and protection against replay attacks. A valid message must only be accepted after the receiver has verified both its origin and its integrity.

Confidentiality + Integrity + Authentication + Replay Protection

2. Cryptographic Requirements

Before exchanging application messages, the communicating devices must establish or derive a session key using a standard cryptographic mechanism. Diffie–Hellman, Elliptic Curve Diffie–Hellman, or a justified pre-shared key scheme may be used. The report must clearly explain how the session key is obtained and how the communicating devices are authenticated.

All text messages must be encrypted before transmission. Authenticated encryption mechanisms such as AES-GCM or ChaCha20-Poly1305 are recommended because they provide confidentiality and integrity within the same construction. If encryption and authentication are implemented separately, the selected design must be technically justified.

The receiving device must reject any message whose ciphertext, authentication information, sender information, or freshness data has been altered. The implementation must also include a mechanism to detect repeated messages. Sequence numbers, counters, timestamps, nonces, or session identifiers may be used for this purpose.

Custom cryptographic algorithms are not permitted. The implementation must use established algorithms and standard cryptographic libraries.

3. Message Format

Each team must define and document the structure of the transmitted packet. The format must include the information required to identify the sender, associate the message with a valid session, provide freshness, carry the encrypted payload, and verify authenticity.

A possible packet format is

$$\boxed{ID_S \parallel SID \parallel SEQ \parallel N \parallel C \parallel TAG}$$

where ID_S identifies the sender, SID identifies the communication session, SEQ represents a sequence number, N is a nonce, C is the ciphertext, and TAG contains the authentication information. This structure is only a reference; each team may define a different format if the design is technically justified.

4. Security Tests

The prototype must be evaluated under normal and adversarial conditions. First, the team must demonstrate successful communication between authorized devices, including session establishment, message encryption, verification, and correct recovery of the plaintext.

The security evaluation must then include deliberate modification of a ciphertext and of its authentication information. In both cases, the receiver must detect the alteration and reject the packet. A previously accepted message must also be captured and transmitted again to verify that the replay protection mechanism works correctly.

The team must additionally attempt to inject a forged packet and to initiate communication from an unauthorized device. These tests must show that the receiver only accepts messages generated within a valid authenticated session.

5. Experimental Evaluation

The technical report must include a basic performance evaluation of the implemented system. At minimum, students must record the plaintext message size, the final protected packet size, the encryption time, the decryption and verification time, and the communication latency.

The analysis should briefly discuss the computational and communication overhead introduced by the cryptographic protection. The objective is not to perform an exhaustive performance study, but to identify the practical cost of securing the communication on the selected IoT platform.

6. Deliverables

The final submission must include the complete source code, a system architecture diagram, a protocol sequence diagram, the specification of the transmitted message format, and a short technical report describing the cryptographic design and the experimental results.

During the final demonstration, the team must show a successful secure message exchange and the required attack scenarios. The implementation must be sufficiently documented so that another student familiar with the selected hardware platform could reproduce the experiment.

7. Evaluation Rubric

Criterion	Weight	Expected Evidence
System Architecture and Protocol Design	15%	Clear architecture, coherent communication flow, and technically justified design decisions.
Secure Session-Key Establishment	15%	Correct establishment or derivation of a session key and appropriate management of cryptographic material.
Confidentiality	15%	Application messages are correctly encrypted before transmission.
Integrity and Message Authentication	15%	Modified or forged packets are detected and rejected.
Device Authentication	10%	Only authorized devices can participate in the secure communication.
Replay Protection	10%	Previously accepted messages cannot be reused as valid new messages.
Security Testing	10%	Correct implementation and interpretation of the required attack tests.
Experimental Evaluation	5%	Appropriate measurement and discussion of cryptographic overhead and communication performance.
Technical Documentation	5%	Clear code, diagrams, protocol description, and reproducible implementation.
Total	100%	

8. Minimum Security Requirement

A solution based only on encryption will not be considered complete. The final prototype must demonstrate confidentiality, authentication, integrity verification, and replay protection as part of a single communication protocol.

Encryption alone $\neq$ Secure Communication

Security claims must be supported by the implemented tests and by the evidence presented during the final demonstration.